

#3D251F
cocap

#3DA2B7
teal

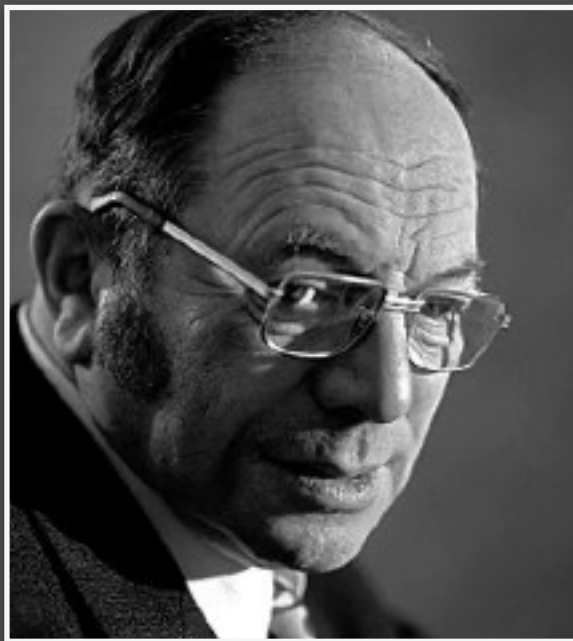
#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Key Takeaway: cost-dependent geometry
enables powerful analysis and optimization



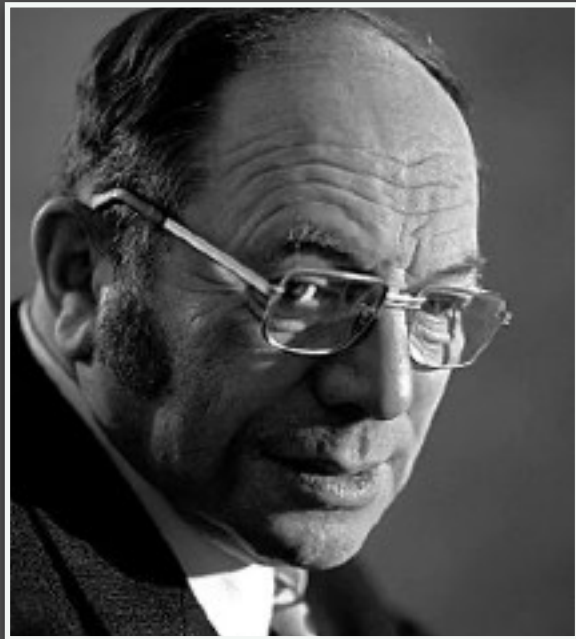
Kantorovich's Problem:

(Continuous version dual)

$$\inf_{f, g : f \oplus g \leq c} \int_X f \, d\alpha + \int_Y g \, d\beta$$

$$W_1(\alpha, \beta) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f d\alpha - \int f d\beta \right)$$

$$W_2^2(\alpha, \beta) = \sup_{\phi \text{ convex}} \left(\int \phi(x) d\alpha(x) + \int \phi^* d\beta(y) \right)$$

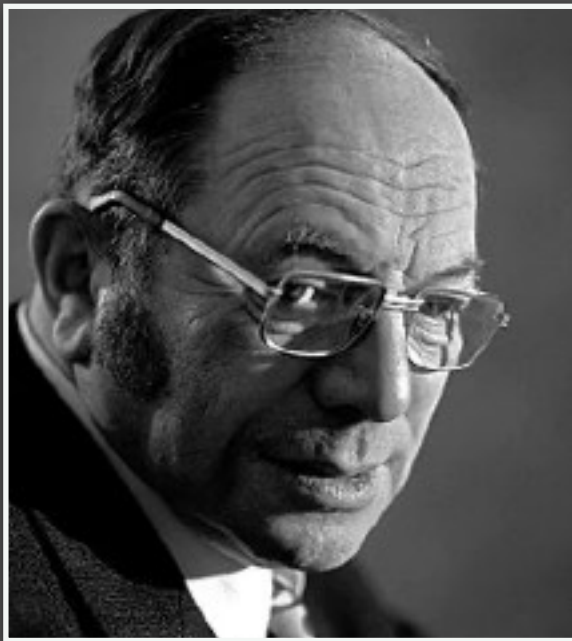


Kantorovich's Problem:

(Continuous version dual w/ c -conjugacy)

$$\sup_{f \in c\text{-CVX}} \int_X f \, d\alpha + \int_Y f^c \, d\beta$$

Make *implicit* interdependence of f, g *explicit*
via c - transforms | conjugacy | convexity



Kantorovich's Problem:

(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$

Sinkhorn Algorithm

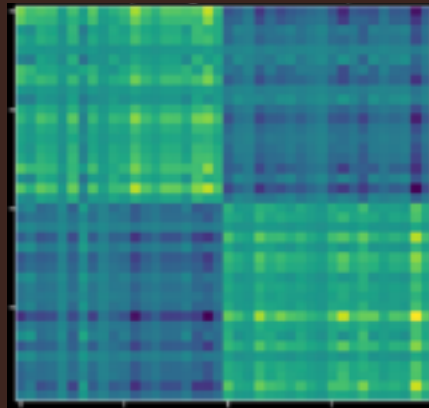
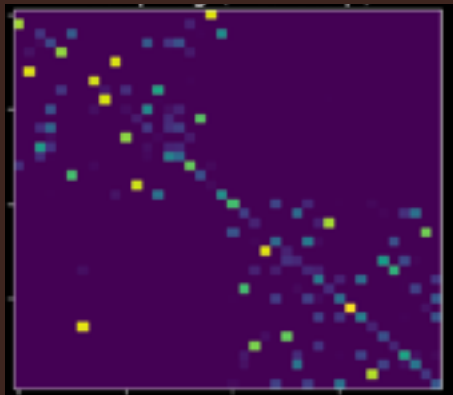
1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$

2. While not converged:

$$\mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K} \mathbf{v}$$

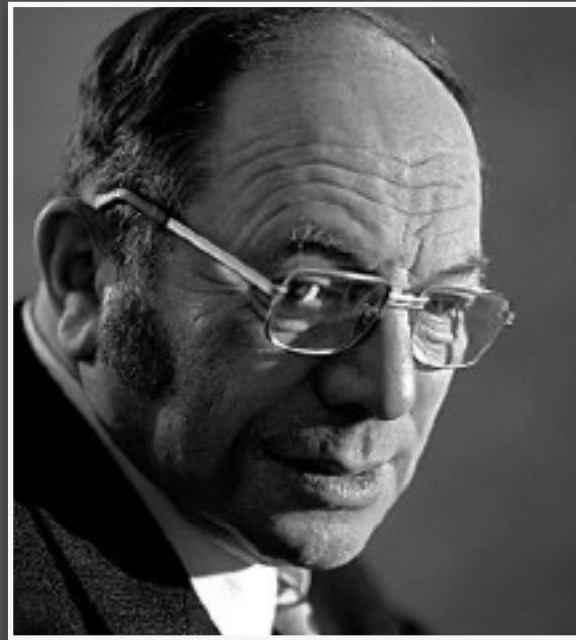
$$\mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^T \mathbf{u}$$

Effect of ε on π



0 $\leftarrow \varepsilon$ $\varepsilon \rightarrow \infty$

Last Time

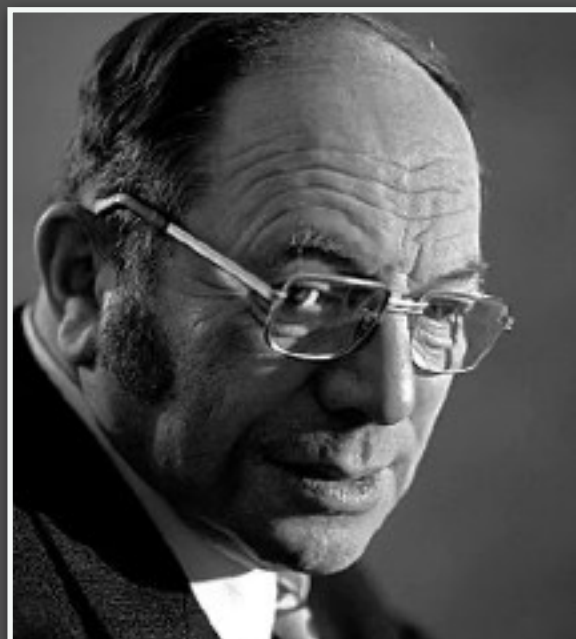


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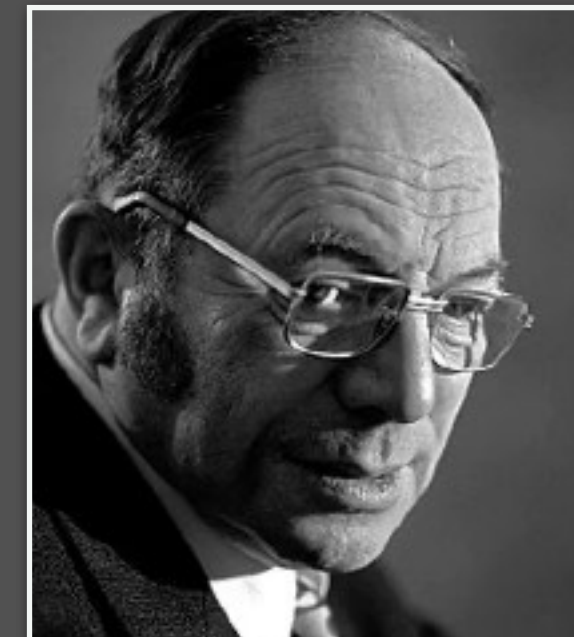
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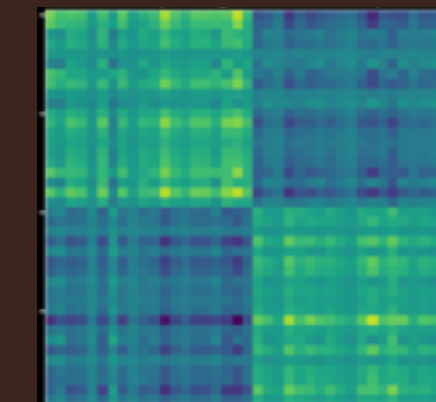
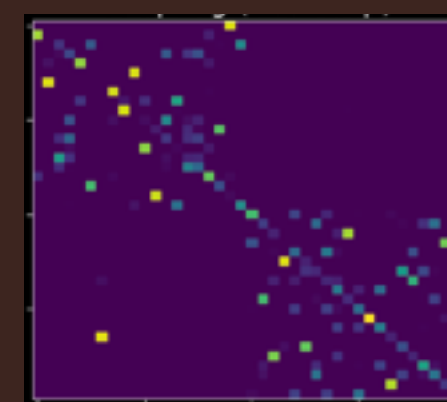


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Today

- **Sinkhorn algorithm:** complexity, convergence, practical considerations
- The dual formulation of Entropy-Regularized OT
- Sinkhorn Divergences